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Estimates of government spending multipliers in Australian data

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ABSTRACT

In this paper, cumulative multipliers for government consumption spending and for government investment spending are estimated by the method of local projections with instrumental variables using quarterly Australian data from 1984Q1 to 2019Q4. This method directly estimates dollar-for-dollar cumulative multipliers. The estimates are for 'average' or non-state-dependent multipliers, and for multipliers which are state dependent. Government consumption spending multipliers have been estimated for Australia but not conditional on an economic state and government investment spending multipliers have not received attention. In this paper, the state-dependent multipliers are estimated for states of high and low unemployment. The peak cumulative multiplier for government investment spending is −0.95 in the non-state-dependent case and −2.54 in the low unemployment state, both for a horizon of six quarters. Government investment spending crowds out private sector activity and has a negligible effect in the high unemployment state. Government consumption spending crowds in (expands) private sector activity only in the high unemployment state and only moderately and has a negligible effect otherwise. The cumulative government consumption spending multiplier in the high unemployment state is 0.94 at six quarters. The results are robust to variations in the baseline specification.

KEYWORDS

Government consumption and investment spending; cumulative multipliers; local projections; instrumental variables; state-dependent multipliers

JEL CLASSIFICATION

C32; E62; H50

I. Introduction

This paper estimates cumulative multipliers for government spending in Australia with quarterly data from 1984Q1 to 2019Q4. Early research estimated 'average' multipliers. These multipliers do not vary over the estimation sample and are non-state dependent. However, recent research has focused on whether government spending multipliers change with economic conditions and are state dependent. This literature estimates state-dependent multipliers conditional on business cycle states (Auerbach and Gorodnichenko 2012, 2013), unemployment states (Owyang, Ramey, and Zubairy 2013; Ramey and Zubairy 2018), private debt overhang (Bernardini and Peersman 2018), monetary policy states (Haug and Power 2022; Ramey and Zubairy 2018), and government debt (Kim 2015). In this paper, we contribute to the Australian literature by investigating whether government spending multipliers are state dependent, specifically, if they depend on whether the economy is in a high or low unemployment state. Carmignani (2022) is the only other study to estimate state-dependent multipliers for Australia, but they are dependent on a political state (an electoral cycle) and not an economic state.

Our other contribution is to estimate cumulative multipliers for the components of Australian government spending for both non-state- and state-dependent specifications. Previous Australian studies have estimated multipliers for either total government spending (Inchauspe 2021) or for government consumption spending (Carmignani 2015, 2022) but not for government investment spending. Boehm (2020) shows that the government consumption multiplier is higher than the government investment multiplier in a conventional dynamic macroeconomic model and finds for a panel of OECD countries that the two multipliers are significantly different from each other. Auerbach and Gorodnichenko (2012, 2013) for the US and Lyu (2023) for New Zealand report different multipliers for government consumption and government investment spending. Haug and Sznajderska (2024) extend the analysis in Ramey and Zubairy (2018) for the US by estimating state-dependent cumulative multipliers for government consumption and government investment spending.

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They find no evidence that government consumption and government investment multipliers are dependent on the unemployment state and that they are statistically different from each other. These results are consistent with the results of Ramey and Zubairy (2018) for total government spending multipliers which they found are not dependent on the unemployment state.

This paper uses Jordà (2005) local projections method (LP) with instrumental variables (IV) developed by Ramey and Zubairy (2018) to estimate state-dependent multipliers for government consumption and government investment spending in Australia. The advantage of their method is that it directly estimates state-dependent, dollar-for-dollar cumulative multipliers. Following Mountford and Uhlig (2009), the multipliers are cumulative to account for the persistence of government spending. The cumulative multipliers are estimated in the LP regressions using the government spending shock of Blanchard and Perotti (2002) as an instrument.[1] In this paper, cumulative multipliers are estimated for two states, a high unemployment state and a low unemployment state. Average (non-state dependent) multipliers are also estimated.

Our main findings are the following. In the high unemployment state, the cumulative multipliers for government consumption spending are positive and less than one and strongly significant for horizons to six quarters. Government consumption spending has a small expansionary effect on private sector activity when the unemployment state is high. The average (non-state dependent) multipliers for government consumption spending are not statistically significant. In the low unemployment state, the cumulative multipliers for government investment spending are negative, greater than one in absolute value and strongly statistically significant after one quarter. Government investment spending crowds out private sector activity when the unemployment state is low. The average multipliers for government investment spending are also negative but less than one in absolute value.

The rest of the paper is organized as follows. Section II describes the econometric methodology and the data. Section III presents the main results. Section IV shows the results for robustness. Section V provides some further discussion and compares the results of the paper to previous Australian studies. Section VI concludes.

II. Methodology

Estimation of multipliers

The econometric approach begins with the specification of the equation for the government spending variable. This equation is:

$$g_t = c + \sum_{i=1}^{n} \alpha_i g_{t-i} + \sum_{i=1}^{n} \beta_i y_{t-i} + \sum_{i=1}^{n} \gamma_i z_{t-i} + \varepsilon_t \tag{1}$$

where g_t is the government spending variable, y_t is the GDP variable and z_t is a vector of control variables with the associated coefficients given in the vector γ_i. Government spending is predetermined since it does not depend on current values of the variables but only their lagged values. Estimation of Equation 1 by OLS gives consistent estimates of the coefficients. The OLS residual is denoted $\widehat{\varepsilon_t}$.

This specification is based on the identifying assumption that government spending does not respond within the quarter to contemporaneous movements in macroeconomic variables. Ramey and Zubairy (2018) refer to the residual in Equation 1 as the Blanchard-Perotti shock. It is the same as the shock to government spending which Blanchard and Perotti (2002) obtained from a Cholesky decomposition of the shocks in a VAR with government spending ordered first.

The equation for the GDP variable is given by:

$$y_t = k + \delta_0 g_t + \sum_{i=1}^{n} \delta_i g_{t-i} + \sum_{i=1}^{n} \eta_i y_{t-i} + \sum_{i=1}^{n} \theta_i z_{t-i} + \upsilon_t \tag{2}$$

The GDP variable depends on contemporaneous government spending and the lags of all the variables. Equation 2 can be estimated consistently by instrumental variable (IV) estimation using the OLS residual from Equation 1, $\widehat{\varepsilon_t}$, as the

[1] Ramey and Zubairy (2018) also use the series US defence news, first constructed by Ramey (2011), as an instrument. However, there is no similar defence news series for Australia.

instrument for g_t. This instrument is uncorrelated with the error term v_t because the recursive specification between the government spending and GDP variables means that ε_t and v_t are orthogonal to each other. This is the approach taken by Haug and Power (2022). However, g_t can be used as its own instrument because g_t is uncorrelated with v_t when ε_t and v_t are orthogonal and OLS can be used to consistently estimate the coefficients in Equation 2. This approach is taken by Ramey and Zubairy (2018) who treat the Blanchard-Perrotti shock as current government spending. We follow their approach, noting that both approaches are equivalent.

The impulse response of the GDP variable at horizon h to a current movement in the government spending variable is given by the estimated value of $\delta_{0,h}$ from the equation:

$$y_{t+h} = k_h + \delta_{0,h} g_t + \sum_{i=1}^{n} \delta_{i,h} g_{t-i} + \sum_{i=1}^{n} \eta_{i,h} y_{t-i} + \sum_{i=1}^{n} \theta_{i,h} z_{t-i} + v_{t+h},$$

for

$$h = 0, 1, 2, \ldots. \tag{3}$$

This method of calculating impulse response functions is an application of Jordà (2005) local projections (LP) method. Mountford and Uhlig (2009) argue that multipliers should be calculated as the sum of the GDP response up to a given horizon divided by the sum of the government spending response up to that horizon as they measure cumulative GDP relative to cumulative government spending following the contemporaneous movement in government spending.

Ramey and Zubairy (2018) estimate the cumulative multiplier and its standard error in one-step. The method is to replace y_{t+h} with the sum of the GDP variable up to horizon h, and to replace g_t with the sum of the government spending variable up to horizon h, in Equation 3 to get:

$$\sum_{i=0}^{h} y_{t+i} = k_h + m_h \sum_{i=0}^{h} g_{t+i} + \sum_{i=1}^{n} \delta_{i,h} g_{t-i} + \sum_{i=1}^{n} \eta_{i,h} y_{t-i} + \sum_{i=1}^{n} \theta_{i,h} z_{t-i} + v_{t+h},$$

for

$$h = 0, 1, 2, \ldots. \tag{4}$$

The estimated value of the coefficient, m_h, is the estimate of the cumulative multiplier for horizon h. They estimate Equation 4 by IV using g_t as the instrument for $\sum_{i=0}^{h} g_{t+i}$, and the lagged variables as their own instruments. The standard error of the multiplier is the standard error of the coefficient, m_h, corrected for serial correlation. Serial correlation in the error term is induced by successive leads of the cumulated GDP variable, i.e. the dependent variable, as the time horizon increases.

They extend this approach to the estimation of cumulative multipliers which depend on an economy's state. For an economy that can be in either state A or state B, the multipliers are obtained by estimating the equation:

$$\begin{aligned}\sum_{i=0}^{h} y_{t+i} = {} & I_{t-1}[k_{A,h} + m_{A,h} \sum_{i=0}^{h} g_{t+i} + \sum_{i=1}^{n} \delta_{A,i,h} g_{t-i} \\ & + \sum_{i=1}^{n} \eta_{A,i,h} y_{t-i} + \sum_{i=1}^{n} \theta_{A,i,h} z_{t-i}] \\ & + (1 - I_{t-1})[k_{B,h} + m_{B,h} \sum_{i=0}^{h} g_{t+i} \\ & + \sum_{i=1}^{n} \delta_{B,i,h} g_{t-i} + \sum_{i=1}^{n} \eta_{B,i,h} y_{t-i} \\ & + \sum_{i=1}^{n} \theta_{B,i,h} z_{t-i}] + v_{t+h}\end{aligned}$$

for

$$h = 0, 1, 2, \ldots \tag{5}$$

where the dummy variable, I_{t-1}, indicates the state of the economy at time $t-1$ with a value of 1 for time periods which correspond to state A and zero for time periods which correspond to state B. The economy's state is dated with a one-period lag to avoid contemporaneous feedback from government spending to the economy's state. All of the coefficients in Equation (5) are allowed to vary according to whether the economy is in state A or state B. Equation (5) is estimated by IV with $(I_{t-1})g_t$ and $(1 - I_{t-1})g_t$ as the instruments for cumulated government spending, $\sum_{i=0}^{h} g_{t+i}$, in states

A and B, respectively. The coefficients $m_{A,h}$ and $m_{B,h}$ are the cumulative government spending multipliers at horizon h for the economy in state A and B, respectively. We report the automatically selected Newy-West HAC standard errors for the cumulative multipliers in both Equation 4 and (5).

Measurement issues with multipliers

It is common to specify the GDP and government spending variables in logs in models to estimate multipliers. Lyu (2023), for example, computes the government spending multiplier as the ratio of the cumulative response of the log of GDP to the cumulative response of the log of government spending multiplied by the sample average of the ratio of GDP to government spending. The sample average scaling factor converts the cumulated log responses into dollar equivalents to obtain dollar-for-dollar multipliers. The use of this ex-post scaling factor can introduce biases in the estimates of the multiplier when the average ratio of GDP to government spending varies widely within subsamples. Importantly, Sims and Wolff (2018) point out that the ratio of GDP to government spending is procyclical which implies that the use of an average GDP to government spending ratio as the scaling factor overestimates the multiplier in recessions and underestimates it in expansions.

Rather than scaling the ratio of the cumulative responses, we follow Ramey and Zubairy (2018) and divide the levels of GDP and government spending by trend GDP prior to the econometric analysis and specify the control variables in levels to obtain dollar-for-dollar multipliers. For this specification, the ratio of the cumulative response of GDP to the cumulative response of government spending over h horizons, following a \$1 increase in current government spending, is $\$m_h$.

Description of the data and variables

The data is quarterly and covers the period 1984Q1 to 2019Q4. The start date corresponds to the beginning of the period of a fully floating exchange rate in Australia while the end date ensures that the sample excludes observations from the COVID-19 pandemic period, which may bias the results. A full description of the data and its sources is provided in the Data Appendix.

The government spending variable is the series for government consumption spending and for government investment spending, respectively. For both series, the spending is by all levels of government together (local, state and commonwealth, collectively known as general government). In the national accounts, total government spending is the sum of government consumption spending, government investment spending and investment spending by public sector corporations. For our sample, government consumption spending averaged 81% of total government spending, government investment spending averaged 12% and investment spending by public sector corporations 7% of the total. We do not consider the latter in our analysis as the time series for public sector corporation investment moves abruptly from time to time.[2] Further public sector corporation investment is not viewed as an instrument of countercyclical fiscal policy as it is determined by the public sector corporations themselves in the course of their market sector activities.

The GDP and government spending series are real series as they are chain volume measures, and they are expressed in per-capita terms by dividing the series by the estimated resident population. They are then divided by the trend in real GDP per capita, as indicated earlier. Following Haug and Power (2022) and Carmignani (2022), the trend in real GDP per capita is obtained by applying the Hodrick-Prescott filter to real GDP per capita with the smoothing parameter set to 1600 which is common practice for quarterly data.

The control variables in z_t are the average tax rate, calculated as the ratio of total nominal taxes to nominal GDP, to control for the financing of government expenditure, the three-month Bank Accepted Bill rate and the inflation rate (based on the basket excluding volatile items) to reflect the stance of monetary policy, and Australia's real trade weighted exchange rate as Australia is a small trade dependent open economy.

[2]This occurs because public sector corporations are often privatized, and new ones come into existence. For example, public sector corporation investment fell by 108% from 1997Q4 to 1998Q1 due to the first stage of the Telstra privatization, it fell by 88% from 1999Q4 to 2000Q1 following the second stage and increased by 63% from 2012Q3 to 2012Q4 following the creation of the National Broadband Network public corporation.

In addition to the non-state-dependent model, Equation 4, we consider the state-dependent model, Equation (5), where the two states of the economy are a high unemployment state and a low unemployment state. The high unemployment state is indicated by the value one and the low unemployment state by the value zero in the series for the dummy variable, i.e. the high unemployment state is state A, and the low unemployment state is state B. In this paper, a high unemployment state occurs when the actual rate of unemployment is above the trend rate of unemployment in quarter t and a low unemployment state occurs when the actual rate is below the trend rate in quarter t. The Hodrick-Prescott filter with smoothing parameter of 1600 is applied to the actual rate of unemployment to extract the trend. Positive deviations of the actual unemployment rate from its trend correspond to quarters of high unemployment and negative deviations to quarters of low unemployment.

Instrument relevance and tests

It is important to test for weak instruments as they may lead to biased estimates of the multipliers. We test the relevance of the instrument used to calculate the cumulative multipliers using the Montiel Olea and Pflueger (2013) first stage effective F-statistic for each state and horizon. An F-statistic below 10 indicates that the instrument is weak (Stock and Yogo 2005). Montiel Olea and Pflueger (2013) propose a stricter threshold when the first stage regression errors are heteroscedastic and serially autocorrelated. This threshold is 19.75 and is for a significance level of 10% in their setup.[3] We use these thresholds to say that the instrument is not weak if the F-statistic is ≥ 10 and that it is strong if the F-statistic is ≥ 19.75.

We test the null hypothesis that the cumulative multipliers are the same in the high and low unemployment states at a given horizon. If the null hypothesis is rejected for all horizons, the multipliers are not state dependent. We report the p-values of the Newy-West HAC based test and the p-values of the Anderson-Rubin (A-R) test, both of which were implemented by Ramey and Zubairy (2018) to test the null. The HAC based test is more powerful provided the instruments are not weak whereas the A-R test is less powerful but robust to weak instruments.

III. Results

Baseline model specification

The model described thus far is the baseline model. It remains to determine the number of lags of the right-hand side variables, i.e. n, in the equations to complete its specification.

Equation 2 was estimated with one to four lags of the right-hand side variables where the government spending variable is, respectively, government consumption and government investment. Table 1 reports the values of the AIC and SBC information criteria and the p-values of the log-likelihood ratio (LR) test for lag length. For government consumption, the AIC selects two lags and the LR test does not reject the null hypothesis of two lags against the alternative of three lags at the 5% significance level (the p-value of the test is 0.10). The SBC selects one lag but that is usually considered too small. Accordingly, we estimate the baseline model for government consumption with $n = 2$ lags in the equations. For government investment, the AIC selects three lags while the SBC selects one lag. The LR test rejects the null hypothesis of two lags against the alternative of three lags (p-value is 0.013) but does not reject the null of

Table 1. Tests for lag length over the effective sample 1985Q1–2019Q4.

	Government consumption			Government investment		
Lag	AIC	SBC	LR	AIC	SBC	LR
1	−746.83	−723.30		−749.95	−726.41	
2	−750.86	−709.67	0.0136	−750.52	−709.33	0.0504
3	−749.48	−690.64	0.1009	−754.61	−695.78	0.0133
4	−741.40	−664.92	0.6871	−746.24	−669.75	0.7271

The entries for LR are the p-values for the log-likelihood ratio test of $n - 1$ lags vs. n lags in Equation 2 in the text. The test statistic is chi-squared distributed with six degrees of freedom as there are six exclusion restrictions under the null, one for each explanatory variable on the right-side of Equation 2. The LR test and the AIC criterion select two lags for government consumption spending and three lags for government investment spending. We chose these lag lengths, notwithstanding that the SBC selects one lag for both.

[3]The threshold at the 5% significance level is 23.11 but we choose the 10% threshold because our sample size is small.

three lags against the alternative of four lags (p-value is 0.727) at the 5% level. As the AIC and the LR test select three lags, we estimate the baseline model for government investment with $n = 3$ lags in the equations.

Results for the non-state-dependent baseline model

Table 2 reports estimates of the cumulative multipliers for government consumption and government investment spending at each horizon for the non-state-dependent baseline model.

For government consumption, the estimated cumulative multipliers are small and are not statistically significant for all horizons. The multipliers are reliably estimated to be indistinguishable from zero as the instrument upon which they are based is strong for horizons of up to five quarters and is not weak for horizons of six and seven quarters. There is no evidence for a government consumption multiplier in our sample for the non-state-dependent case.

Table 2. Estimates of cumulative government spending multipliers in the non-state-dependent baseline model.

	Government Spending			
	Government Consumption		Government Investment	
Horizon	Multiplier	F-statistic	Multiplier	F-statistic
0	0.083	–	0.118	–
	(0.136)		(0.214)	
1	0.216	203.31	−0.057	146.53
	(0.172)		(0.241)	
2	0.094	59.83	−0.457*	46.25
	(0.186)		(0.276)	
3	−0.092	33.90	−0.630**	39.38
	(0.217)		(0.291)	
4	−0.209	22.18	−0.742***	41.21
	(0.232)		(0.279)	
5	−0.161	20.90	−0.882***	29.34
	(0.226)		(0.302)	
6	0.001	16.17	−0.948***	25.11
	(0.201)		(0.312)	
7	0.046	11.97	−0.811***	23.91
	(0.211)		(0.283)	
8	0.029	8.57	−0.765***	19.94
	(0.228)		(0.271)	
9	0.083	7.79	−0.719***	17.57
	(0.266)		(0.266)	
10	0.066	6.99	−0.669**	15.95
	(0.270)		(0.263)	
11	0.146	6.49	−0.636**	12.12
	(0.290)		(0.244)	
12	0.176	6.32	−0.533***	11.89
	(0.288)		(0.192)	

The horizon is in quarters. The values in parentheses under the multipliers are Newey-West HAC adjusted standard errors. The F-statistic is the Montiel Olea and Pflueger (2013) first stage effective F-statistic for instrument relevance. The effective F-statistic is from the regression of the sum of government spending to horizon h on the instrument at t, g_t, and all the controls from the second stage. When the F-statistic is ≥10, the instrument is not weak, and when it is ≥19.75, the instrument is strong. The F-statistic cannot be computed at horizon 0 because the instrument and the variable it is the instrument for are the same, namely g_t. The asterisks ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

For government investment, the estimated cumulative multipliers are negative, less than one (in absolute value) and significant from the two-quarter horizon. The multipliers range from −0.46 at the two-quarter horizon to −0.95 at the six-quarter horizon and by twelve quarters the multiplier is −0.53. The multiplier of −0.95, the largest in absolute terms, means that following an increase of $1 in current government investment spending, the ratio of the cumulative response of GDP to the cumulative response of government investment falls by $0.95 over the next six quarters. This fall is statistically significant at the 1% level. The multipliers are reliably estimated because the instruments are strong for horizons to seven quarters and are not weak for subsequent horizons. In the non-state-dependent baseline model, there is strong evidence that the multipliers for government investment spending are negative, less than one in absolute terms, and are statistically significant. The negative multipliers imply that following a current rise in government investment spending, there is a crowding out of private sector spending so that cumulative GDP falls over subsequent horizons.

Results for the state-dependent baseline model

Table 3 reports estimates of the cumulative multipliers for government consumption spending in the state of high and low unemployment, respectively, for the baseline model. For the high unemployment state, the cumulative multipliers are positive and generally less than one and are statistically significant at either the 1% or 5% level for horizons up to six quarters. The multipliers are estimated reliably since the instruments are strong for horizons to two quarters and are not weak for horizons from three to six quarters. For longer horizons, the multipliers are statistically significant at the 10% level but the instruments are weak. Among the multipliers that are significant and based on an instrument which is either strong or not weak, the largest is 1.09 at the two-quarter horizon. This multiplier says that the ratio of the cumulative response of GDP to the

Table 3. Estimates of cumulative government consumption multipliers in the state-dependent baseline model.

	Government Consumption Spending					
	High Unemployment State		Low Unemployment State		p-value for hypothesis test of no difference	
Horizon	Multiplier	F-statistic	Multiplier	F-statistic	HAC	A-R
0	0.538***	–	−0.002	–	0.018	–
	(0.149)		(0.233)			
1	0.921***	132.16	0.197	126.52	0.042	0.120
	(0.251)		(0.320)			
2	1.088***	31.79	0.042	46.58	0.012	0.086
	(0.330)		(0.330)			
3	0.856**	13.71	−0.143	32.33	0.014	0.087
	(0.333)		(0.542)			
4	0.735**	11.13	−0.360	34.43	0.008	0.070
	(0.353)		(0.327)			
5	0.763**	13.50	−0.473	23.78	0.008	0.076
	(0.334)		(0.368)			
6	0.938**	12.45	−0.489	15.06	0.007	0.072
	(0.390)		(0.329)			
7	1.210*	7.02	−0.559**	13.60	0.015	0.061
	(0.619)		(0.266)			
8	1.058*	6.68	−0.597**	7.44	0.021	0.075
	(0.572)		(0.285)			
9	0.982*	6.94	−0.556*	6.31	0.023	0.091
	(0.515)		(0.312)			
10	0.769*	7.65	−0.481	5.85	0.045	0.144
	(0.404)		(0.389)			
11	0.695*	6.18	−0.433	4.32	0.066	0.161
	(0.395)		(0.397)			
12	0.552*	5.84	−0.414	3.30	0.075	0.177
	(0.313)		(0.394)			

The values in parentheses under the multipliers are Newey-West HAC adjusted standard errors and the F-statistic is the Montiel Olea and Pflueger (2013) first stage effective F-statistic for instrument relevance. See the notes to Table 2. The HAC and Anderson-Rubin (A-R) p-values are for the test of the null hypothesis that the multipliers are the same in the high and low unemployment state for horizon *h*. The HAC p-values are based on Newey-West standard errors adjusted for serial correlation and heteroscedasticity and the A-R p-values are robust to weak instruments. The F-statistic and the A-R p-value cannot be computed at horizon 0 because the instrument and the variable it is instrumenting for are the same. The asterisks ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

cumulative response of government consumption over the next two quarters is $1.09 following an immediate increase of $1 in government consumption spending. There is reliable evidence for a cumulative multiplier less than or close to unity for government consumption spending in the high unemployment state for horizons up to six-quarters.

For the low unemployment state, the cumulative multipliers for government consumption spending are not statistically significant for horizons out to six quarters. They are statistically significant and negative at horizons of seven, eight and nine quarters but for the latter two horizons the instrument is weak. At longer horizons, the multipliers are statistically insignificant, and the instruments are weak. There is no evidence of a multiplier effect of government consumption spending in the low unemployment state, except possibly at seven quarters.

The last two columns of Table 3 report the p-values for the Newey-West HAC test statistic and for the A-R test statistic, respectively, for the null hypothesis that the cumulative multipliers are the same in both states at a given horizon. The HAC test rejects the null hypothesis of identical multipliers in both states at the 5% level for horizons of one to six quarters and the A-R test rejects the null hypothesis for horizons of two to six quarters at the 10% level. These results are not surprising since the multipliers in the high unemployment state are statistically significant while those in the low unemployment state are not significant for these horizons.

Table 4 reports the estimated cumulative multipliers for government investment spending in the two states for the baseline model. For the high

Table 4. Estimates of cumulative government investment multipliers in the state-dependent baseline model.

	Government Investment Spending					
	High Unemployment State		Low Unemployment State		p-value for hypothesis test of no difference	
Horizon	Multiplier	F-statistic	Multiplier	F-statistic	HAC	A-R
0	0.595	–	−0.034	–	0.197	–
	(0.413)		(0.327)			
1	0.566	288.86	−0.477	172.09	0.018	0.052
	(0.404)		(0.419)			
2	0.436	64.93	−1.411***	69.79	0.002	0.037
	(0.457)		(0.525)			
3	0.448	53.04	−1.582**	69.77	0.009	0.045
	(0.395)		(0.707)			
4	0.359	34.52	−2.016**	28.85	0.003	0.022
	(0.340)		(0.805)			
5	0.183	24.02	−2.436***	19.13	0.004	0.033
	(0.311)		(0.862)			
6	0.038	19.30	−2.539***	15.68	0.008	0.054
	(0.311)		(0.921)			
7	0.072	22.39	−2.251**	15.01	0.014	0.062
	(0.308)		(0.874)			
8	0.158	19.16	−2.016**	12.44	0.015	0.062
	(0.288)		(0.789)			
9	0.252	17.29	−1.735**	9.65	0.017	0.053
	(0.269)		(0.728)			
10	0.285	17.95	−1.547**	7.23	0.017	0.056
	(0.256)		(0.661)			
11	0.353	14.36	−1.358**	5.87	0.010	0.059
	(0.294)		(0.519)			
12	0.390	13.06	−1.028***	6.11	0.005	0.061
	(0.298)		(0.371)			

See the notes to Table 3.

unemployment state, the estimated multipliers are positive and less than one and are not statistically significant. The multipliers are reliably estimated as not significant because they are estimated with an instrument which is either strong or not weak. There is no evidence for a multiplier effect of government investment spending in the high unemployment state.

For the low unemployment state, the estimated multipliers are negative. They are larger than one (in absolute value) for horizons of two and three quarters, and larger than two (in absolute value) for horizons from four to eight quarters. For these horizons, the multipliers are statistically significant at either the 1% or 5% level and estimated with an instrument which is either strong or not weak. The peak multiplier of −2.54 says that in response to a rise of $1 in current government investment spending in the low unemployment state, the ratio of the cumulative response of GDP to the cumulative response of government investment spending over six quarters falls by $2.54. This is much larger than the fall of $0.95 that was found for government investment at the six-quarter horizon in the non-state-dependent baseline model. Our results show that government investment spending crowds out private sector activity by much more in the low unemployment state than on average (non-state-dependent case).

The last two columns of Table 4 show, respectively, the p-values for the Newey-West HAC and A-R test statistics for the null hypothesis that the cumulative multipliers are the same in both states. The HAC test rejects the null at the 5% level for all horizons and, importantly, the A-R test rejects the null at the 10% level for horizons where the instrument is weak. These results are not surprising since the multipliers in the low unemployment state are negative and large (in absolute value) and statistically significant while the multipliers in the high unemployment state are not statistically different from zero.

To conclude this section, the multipliers for government consumption and investment spending are dependent on the unemployment state. Government consumption spending has a small expansionary effect on private sector activity in the high unemployment state while government investment spending crowds out private sector activity in the low unemployment state.

IV. Robustness

In this section, we investigate the sensitivity of the results from the baseline model to variations in the modelling choices one at a time.

Sensitivity to lag length

Table 5 reports the cumulative multipliers for government consumption spending in the baseline model when the number of lags is four and six. The results are very similar to those reported in Table 3 for two lags. In particular, the finding of significant positive and smaller than one cumulative government consumption multipliers in the high unemployment state is robust across the lag lengths.

Table 6 reports the cumulative multipliers for government investment spending in the four and six lag specifications. The results are very similar to those reported in Table 4 for three lags, especially for the multipliers in the low unemployment state. For the high unemployment state, there are some differences, however. In the four-lag model, some of cumulative high unemployment state multipliers are significantly positive and less than one. In the six-lag model they are significantly positive and larger than one, although they are generally based on a weak instrument for horizons of more than four quarters.[4]

The finding that the cumulative government investment multipliers in the low unemployment state are negative, larger than one (in absolute value) and are statistically significant is robust across lag lengths. Somewhat less robust to variations in lag length is the finding that the government investment multipliers are not statistically significant in the high unemployment state. The non-state-dependent multipliers are robust across lag lengths.

[4]This finding does not appear in the model with eight lags. In that model, the multipliers in the high unemployment state are statistically significant out to horizons of four quarters but are less than one and based on an instrument which is weak. The F-statistic for instrument relevance is less than five at all horizons.

Table 5. Estimates of cumulative government consumption multipliers in the baseline models for four and six lags, respectively.

	Government Consumption Spending					
	Non-State-Dependent Case		High Unemployment State		Low Unemployment State	
Horizon	Multiplier	F-statistic	Multiplier	F-statistic	Multiplier	F-statistic
		Lags = 4				
2	0.177	40.09	0.964**	23.57	−0.075	41.09
	(0.191)		(0.387)		(0.347)	
4	−0.121	16.81	0.541*	14.66	−0.309	33.08
	(0.252)		(0.304)		(0.397)	
6	0.122	12.25	0.527*	22.63	−0.346	13.78
	(0.196)		(0.270)		(0.386)	
8	0.110	7.13	0.383	8.98	−0.501*	6.47
	(0.182)		(0.321)		(0.286)	
10	0.081	5.72	0.216	8.83	−0.314	1.78
	(0.174)		(0.235)		(0.461)	
12	0.132	5.00	0.144	6.32	−0.270	1.22
	(0.188)		(0.248)		(0.412)	
		Lags = 6				
2	−0.094	43.73	0.983**	50.20	0.020	62.30
	(0.246)		(0.464)		(0.384)	
4	−0.374	19.01	0.591	28.88	−0.016	73.74
	(0.326)		(0.500)		(0.362)	
6	−0.068	13.77	0.579	29.52	0.254	23.87
	(0.229)		(0.481)		(0.346)	
8	−0.076	8.36	0.258	15.68	0.191	11.93
	(0.176)		(0.404)		(0.286)	
10	−0.115	6.78	0.132	13.11	0.834*	5.40
	(0.184)		(0.398)		(0.431)	
12	−0.052	5.33	0.259	8.52	0.560	3.17
	(0.177)		(0.388)		(0.413)	

See notes to Table 2.

Table 6. Estimates of cumulative government investment multipliers in the baseline models for four and six lags, respectively.

	Government Investment Spending					
	Non-State-Dependent Case		High Unemployment State		Low Unemployment State	
Horizon	Multiplier	F-statistic	Multiplier	F-statistic	Multiplier	F-statistic
		Lags = 4				
2	−0.529*	64.09	0.561**	61.48	−1.453**	45.05
	(0.306)		(0.267)		(0.564)	
4	−0.775***	67.41	0.389	37.70	−1.884**	20.23
	(0.295)		(0.252)		(0.751)	
6	−0.913***	42.26	0.136	19.36	−2.089**	14.48
	(0.303)		(0.302)		(0.894)	
8	−0.709**	34.16	0.282	21.47	−1.616**	13.35
	(0.281)		(0.214)		(0.809)	
10	−0.617**	26.97	0.518**	18.43	−1.373**	7.93
	(0.283)		(0.243)		(0.773)	
12	−0.493**	20.93	0.587*	12.35	−1.068*	6.44
	(0.237)		(0.310)		(0.498)	
		Lags = 6				
2	−0.454	93.14	1.354*	40.99	−1.590***	25.38
	(0.450)		(0.710)		(0.518)	
4	−0.781*	102.48	1.154**	13.55	−1.942***	24.24
	(0.429)		(0.556)		(0.427)	
6	−0.784**	53.67	1.206*	5.86	−1.946***	19.46
	(0.390)		(0.692)		(0.535)	
8	−0.483*	46.66	1.556*	13.81	−1.315***	11.06
	(0.288)		(0.814)		(0.495)	
10	−0.436*	33.78	1.799*	9.19	−1.021**	6.00
	(0.246)		(0.977)		(0.479)	
12	−0.290	27.43	2.891	3.74	−0.791***	4.80
	(0.179)		(1.796)		(0.286)	

See notes to Table 2.

Table 7. Estimates of cumulative government spending multipliers in baseline model: world industrial production index included in controls.

	Government Consumption Spending					
	Non-State-Dependent Case		High Unemployment State		Low Unemployment State	
Horizon	Multiplier	F-statistic	Multiplier	F-statistic	Multiplier	F-statistic
2	0.116	72.63	1.280***	35.48	0.263	63.94
	(0.189)		(0.310)		(0.387)	
4	−0.227	27.36	0.920***	13.66	−0.143	64.61
	(0.249)		(0.304)		(0.392)	
6	−0.054	16.73	1.041***	15.14	−0.326	27.51
	(0.184)		(0.311)		(0.367)	
8	−0.071	7.76	1.069**	7.63	−0.512	13.86
	(0.216)		(0.455)		(0.360)	
10	−0.072	5.09	0.750**	9.67	−0.635	4.77
	(0.283)		(0.329)		(0.583)	
12	−0.018	3.02	0.500*	5.47	−0.551	3.35
	(0.345)		(0.293)		(0.520)	
			Government Investment Spending			
2	−0.317	51.71	0.959*	55.12	−1.396***	72.95
	(0.314)		(0.511)		(0.455)	
4	−0.571*	69.70	0.831*	36.43	−1.822**	25.23
	(0.293)		(0.443)		(0.849)	
6	−0.715***	40.89	0.454	26.83	−2.292**	25.20
	(0.258)		(0.406)		(0.998)	
8	−0.513***	29.60	0.564	23.41	−1.670**	36.83
	(0.182)		(0.389)		(0.781)	
10	−0.461**	20.91	0.641*	18.13	−1.116	11.18
	(0.195)		(0.339)		(0.830)	
12	−0.374**	12.66	0.690*	12.29	−0.506	3.59
	(0.173)		(0.394)		(0.694)	

See notes to Table 2.

Inclusion of an additional control variable

The next variation is to include the world industrial production index, constructed by Baumeister and Hamilton (2019), in the vector of controls. This is an exogenous variable that controls for the impact of global economic activity and is likely to be important for a small trade dependent economy like Australia.

Table 7 reports the cumulative multipliers for government consumption and government investment spending when the world industrial production index is included in the vector of controls in the baseline models. In the high unemployment state, the government consumption multipliers are positive, near or larger than one, statistically significant and based on an instrument which is either strong or not weak for horizons up to six quarters. These multipliers are somewhat higher than those reported in Table 3 for the baseline model.

For government investment, the multipliers in the high unemployment state are positive and less than one, and unlike in the baseline model (Table 4) are significant at the 10% level up to horizons of four quarters. In the low unemployment state, the multipliers for government investment spending are negative and larger than one in absolute value, and statistically significant for horizons from two to eight quarters just as in Table 4. A comparison of the entries in Tables 2–4 with the corresponding entries in Table 7 shows that the results change little when the vector of controls includes the world industrial production index.

An alternate normalization of GDP and government spending variables

The next variation we consider is to divide the GDP and government spending variables not by the HP trend but rather by the fourth-order polynomial trend in real GDP per capita.[5]

Table 8 reports the cumulative multipliers for government consumption and government investment

[5]This is obtained by regressing real GDP per capita on a fourth-order polynomial in time. The predicted values from the regression are the fourth-order polynomial trend values in real GDP per capita.

Table 8. Estimates of cumulative government spending multipliers in the baseline model: polynomial trend normalization.

	Government Consumption Spending					
	Non-State-Dependent Case		High Unemployment State		Low Unemployment State	
Horizon	Multiplier	F-statistic	Multiplier	F-statistic	Multiplier	F-statistic
2	0.421*	45.83	1.563***	39.39	0.237	54.95
	(0.165)		(0.363)		(0.320)	
4	0.367	16.10	1.322***	14.10	0.101	31.55
	(0.281)		(0.399)		(0.442)	
6	0.683	11.38	1.345***	14.98	0.299	15.70
	(0.450)		(0.381)		(0.619)	
8	0.813	6.11	1.414***	6.70	0.372	8.44
	(0.612)		(0.508)		(0.700)	
10	0.879	4.82	1.142***	6.86	0.952	5.71
	(0.699)		(0.366)		(1.075)	
12	0.845	4.09	0.608	4.57	0.854	3.32
	(0.898)		(0.447)		(1.094)	
			Government Investment Spending			
2	−0.353	47.60	0.527	64.31	−1.452**	54.63
	(0.345)		(0.405)		(0.625)	
4	−0.599	43.37	0.601	33.54	−2.073**	28.03
	(0.402)		(0.422)		(0.906)	
6	−0.724	35.99	0.456	25.69	−2.464**	14.36
	(0.520)		(0.482)		(1.229)	
8	−0.467	26.57	0.717	25.04	−2.002*	11.54
	(0.480)		(0.514)		(1.011)	
10	−0.281	22.26	0.948*	23.41	−1.421*	6.80
	(0.454)		(0.569)		(0.827)	
12	−0.127	14.49	1.258*	17.23	−0.921	5.50
	(0.459)		(0.727)		(0.687)	

See notes to Table 2. GDP and government consumption and government investment spending are normalized by a fourth-order polynomial trend in real GDP per-capita.

spending for this normalization in the baseline model. In the high unemployment state, the multipliers for government consumption spending are positive, larger than one and statistically significant, and based on an instrument that is not weak, for horizons to six quarters. These multipliers are larger than those in the baseline model (Table 3) which are generally less than one but are also significant and based on an instrument which is not weak. In the low unemployment state, the government consumption multipliers are not statistically significant which is generally consistent with the results for the baseline model reported in Table 3.

The cumulative multipliers for government investment spending in the low unemployment state are larger than one and for some horizons larger than two (in absolute value) and are significant as in the baseline model (Table 4). They are generally not statistically significant in the high unemployment state. The non-state-dependent government investment multipliers are negative and less than one in absolute value but unlike in the baseline model (Table 2) they are not statistically significant. Overall though the results from the baseline model are robust to this normalization.

An alternate statistical classification of unemployment states

As a robustness check, we define the states, not with reference to the HP trend unemployment rate, but with reference to a four-quarter centred moving average of the unemployment rate as the trend. The cumulative multipliers for this variation are shown in Table 9 for both government consumption and government investment spending. These multipliers are very similar to those reported in Table 3 for government consumption and in Table 4 for government investment spending. In particular, the government consumption multipliers in the high unemployment state are positive, significant and increase from below one to just above one by six quarters. The government investment multipliers in the low unemployment state are negative and larger than one (in absolute

Table 9. Estimates of cumulative government spending multipliers in the baseline model: unemployment states based on a moving average of unemployment rate.

	Government Consumption Spending					
	High Unemployment State		Low Unemployment State		p-value for hypothesis test of no difference	
Horizon	Multiplier	F-statistic	Multiplier	F-statistic	HAC	A-R
2	0.744*	42.88	−0.192	117.13	0.138	0.218
	(0.396)		(0.308)			
4	0.741*	13.75	−0.817*	47.36	0.058	0.109
	(0.437)		(0.482)			
6	1.038**	11.67	−0.940**	31.85	0.029	0.079
	(0.476)		(0.468)			
8	0.696	5.84	−0.918	11.46	0.091	0.132
	(0.488)		(0.559)			
10	0.182	5.45	−0.662	6.67	0.259	0.288
	(0.298)		(0.554)			
12	−0.307	8.44	−0.290	4.35	0.978	0.978
	(0.445)		(0.388)			
	Government Investment Spending					
2	0.082	32.21	−1.034**	58.29	0.037	0.074
	(0.331)		(0.397)			
4	0.198	32.55	−1.926***	20.55	0.006	0.035
	(0.445)		(0.658)			
6	0.211	23.96	−2.185**	10.36	0.018	0.047
	(0.476)		(0.862)			
8	0.394	20.33	−1.777**	9.13	0.042	0.062
	(0.482)		(0.796)			
10	0.594	12.78	−1.375**	7.71	0.051	0.061
	(0.460)		(0.692)			
12	0.947	5.35	−0.982**	6.08	0.039	0.071
	(0.672)		(0.446)			

See notes to Table 3. A high (low) unemployment state occurs when the actual unemployment rate is above (below) the 4-quarter centred moving average unemployment rate.

terms), and statistically significant. The baseline results are robust to this way of obtaining the unemployment states.

Total government spending

We now replace the component of government spending in the baseline model with total government spending, defined as the sum of government consumption and government investment spending.[6] Table 10 reports the cumulative multipliers for total government spending for the baseline model with three lags. The non-state-dependent multipliers are generally negative and small and except at intermediate horizons are not statistically significant. The cumulative total government spending multipliers in the high unemployment state are positive, less than one, based on strong instruments and statistically significant for horizons up to three quarters. In the low unemployment state, the cumulative total spending multipliers are negative and statistically significant for horizons from two to eight quarters. They are less than one at horizons of two and three quarters and larger than one (both in absolute terms) for subsequent horizons. Overall, the total government spending multipliers are a composite of the multipliers for the two components of government spending.

V. Discussion

Predictability of government spending shocks

Ramey and Zubairy (2018) point out that the Blanchard-Perotti shock, or, equivalently, the shock in Equation 1, is sensitive to measurement error in the series for government spending and is likely to be not entirely unpredictable. Recall that we implemented the Blanchard and Perotti identification assumption in the local projection framework by setting $\hat{\varepsilon}_t$, the instrument, equal to g_t, the government spending variable. This is valid when the set of lagged variables in the local projections

[6]This definition of total government spending is different from the national accounts definition because it does not include investment spending by public sector corporations.

Table 10. Estimates of cumulative total government spending multipliers in the baseline model.

	Total Government Spending					
	Non-State-Dependent Case		High Unemployment State		Low Unemployment State	
Horizon	Multiplier	F-statistic	Multiplier	F-statistic	Multiplier	F-statistic
0	0.193	–	0.674***	–	0.074	–
	(0.140)		(0.224)		(0.161)	
1	0.224	793.08	0.797***	166.61	0.090	276.17
	(0.179)		(0.227)		(0.240)	
2	−0.030	152.56	0.907***	57.93	−0.443*	87.05
	(0.211)		(0.282)		(0.249)	
3	−0.285	79.73	0.669**	71.50	−0.785**	38.16
	(0.219)		(0.288)		(0.336)	
4	−0.450	75.77	0.457	41.83	−1.149**	25.49
	(0.286)		(0.308)		(0.517)	
6	−0.466**	27.07	0.233	35.08	−1.412***	13.19
	(0.214)		(0.299)		(0.444)	
8	−0.406**	10.00	0.225	17.43	−1.395**	7.19
	(0.188)		(0.297)		(0.568)	
10	−0.366**	4.98	0.181	14.20	−1.316	2.42
	(0.180)		(0.243)		(0.815)	
12	−0.252	2.80	0.378	3.64	−1.002	1.67
	(0.172)		(0.401)		(0.634)	

See notes to Table 2. Total government spending is the sum of government consumption spending and government investment spending. The number of lags in the baseline model is three.

model (Equation 4 or Equation (5)) contains the set of lagged variables in Equation 1, the equation for the fiscal policy rule.

We investigated whether the set of lagged variables on the right-hand side of Equation 1, can predict future values of the Blanchard-Perotti government consumption and government investment spending shocks. Specifically, we regressed $\hat{\varepsilon}_{t+h}$ for horizons $h = 1, 2, \ldots, 12$, on the information set of the baseline model at time *t-1*, i.e. on the lags of g_t, y_t and the control variables in z_t. Consistent with the baseline specification, we selected two lags of the variables in the predictive regressions for the shock to government consumption spending and three lags for the shock to government investment spending. In the regressions for both shocks, the coefficients on all the predictive variables taken together were not jointly statistically significant at the 5% level for all forecast horizons.

Further, we found that the coefficients on the lags of each predictive variable were not jointly statistically significant at the 5% level in the regressions except in the following small number of cases. The coefficients on the lags of the real exchange rate were jointly statistically significant at the 5% level in the regression for the government investment spending shock at horizons of three and four quarters, and the lags of government consumption spending were jointly significant in the regression for the government consumption spending shock at the four-quarter horizon. Overall, the evidence suggests that there is little predictable variation in the Blanchard and Perotti government consumption and investment spending shocks, conditional on the baseline model's information set, and that they are likely to be unanticipated.

Relation to VARs

In this paper, impulse responses and cumulative multipliers are estimated by local projections (LP) with instrumental variables (IV). Playborg-Møller and Wolf (2021) prove the equivalence of impulse responses from VARs and LPs for unrestricted lag structures for both linear and non-linear data generating processes. They say that, in finite samples, LPs with p lags for the controls (including g_t and y_t in our case) and VARs with p lags 'agree approximately at impulse response horizons $h \leq p$' (Playborg-Møller and Wolf 2021, 974–975). This means the impulse responses and multipliers from VARs and LPs will agree at short and medium-term horizons provided both methods use the same number of sufficient lags.

In the previous section, we found that the results in the baseline model for government consumption and government investment spending were robust when the number of lags was increased to six.

Given the findings of Playborg-Møller and Wolf (2021), we would expect the results from VARs (both linear and state dependent) of orders to six lags to agree with the results from our LP approach.

Comparison to previous Australian studies

Carmignani (2015) estimated a structural vector-autoregressive (SVAR) model with Australian data for the sample 1978Q1-2014Q4 and obtained estimates of the cumulative multipliers for government consumption spending. In the SVAR government consumption spending is ordered first so it does not depend on contemporaneous variables, just as in our case. The cumulative multipliers were found to be positive and larger than one. At the four-quarter horizon, the cumulative multiplier is 1.2 and is statistically significant. The relevant comparison is with the results we obtained for the non-state-dependent multipliers because the impulse responses in Carmignani's SVAR are not state-dependent. [7] We found the non-state-dependent cumulative government consumption multipliers to be statistically insignificant at all horizons. One possible explanation for the discrepancy between both results is that the impulse responses in the SVAR are scaled to convert elasticities to dollar terms and that may bias estimates of the multipliers as discussed previously.

Inchauspe (2021) estimated a SVAR along the lines of Mountford and Uhlig (2009) with Australian data over the sample 1980Q1-2019Q1 to obtain estimates of the cumulative multipliers for total government spending. The SVAR includes many variables that account for monetary and fiscal policy and the business cycle, and the shocks are identified by a combination of sign and zero restrictions on the impulse responses. A shock to total government spending under the restriction that it does not have a contemporaneous effect on government tax revenues is defined as a deficit financed expenditure shock in the SVAR. The paper finds that the cumulative multiplier for total government spending in the deficit financed case is 0.42 at the one-quarter horizon, 0.03 at four quarters, −0.13 at six quarters and 0.14 at twelve quarters (Table 2, p. 3998) and all are statistically insignificant (Table 4, p. 4000). These can loosely be compared with the non-state-dependent cumulative total government spending multipliers reported in Table 10 which are also small and statistically insignificant out to four quarters. Our multiplier is also negative at six quarters (−0.47) but is statistically significant.

Makin and Narayan (2011) infer the likely size of the multiplier indirectly by evaluating the response of private saving to changes in public saving. They found that 75% to near 100% of the change in public saving is offset by changes in private savings for the period 1980–2008 implying a small or near zero total government expenditure multiplier in Australia. This is consistent with our findings for the non-state-dependent case out to four quarters (Table 10) and with the findings of Inchauspe (2021).

Carmignani (2022) applied the method of Ramey and Zubairy (2018) to estimate cumulative government consumption spending multipliers for Australia for the period 1960Q1-2021Q4. The multipliers depend, not on an economic state, but a political state. A quarter is classified as either an electoral state quarter or as a non-electoral state quarter. The former is associated with a forthcoming national election and the latter is not. For electoral states, the cumulative multipliers for government consumption are reported for horizons of three- and seven-quarters as 3.62 and 2.48, respectively, and are statistically significant while for non-electoral states, these multipliers are negative and very close to zero and are not statistically significant (see Carmignani 2022, Table 2, p. 942). These multipliers are not comparable to ours because they condition on a political state and not an unemployment state. However, for the high unemployment state, we find statistically significant cumulative government consumption multipliers, albeit less than one. Carmignani (2022, Table 1, p.942) also reports the non-state-dependent impact multiplier for government consumption spending as 0.098, which is comparable to our estimate of 0.083 shown in Table 2 and both are not statistically significant.

VI. Conclusion

This paper applies the method of local projections with instrumental variables in a state-dependent framework to directly estimate the

[7]The SVAR of Shaheen and Turner (2020) for the UK is non-linear and there the impulse responses to fiscal policy shocks are state dependent.

cumulative multipliers for government consumption and government investment spending for Australia. We find that government investment spending crowds out private sector activity on average (the non-state-dependent case) and that the crowding out is much stronger when the economy is in the low unemployment state. Government consumption spending has a small simulative effect on private sector activity in the high unemployment state but otherwise has little effect. These finding are robust to variations in the baseline model.

There are some issues which are left for future research. One is to investigate whether the cumulative multipliers depend on states of the economy other than the unemployment state, for example, states of high and low government or private debt, and states of tight and loose monetary policy. Another is to find an alternative instrument to the Blanchard and Perotti (2002) shock as a check on the robustness of the results. However, it is difficult in practice to obtain extraneous information about government spending which is not related to current economic conditions. A third is to investigate the multiplier effects of government consumption and investment spending on the components of private sector activity.

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Data availability statement

The authors confirm that the data supporting the findings of this study are available within the article or its supplementary materials.

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Data Appendix

Government consumption spending: General Government Final Consumption Expenditure, chain volume measure, $ million, seasonally adjusted. Series ID: A2304080V. Source: Australian Bureau of Statistics, Catalogue number 5206.0, Table 2.

Government investment spending: General Government Gross Fixed Capital Formation, chain volume measure, $ million, seasonally adjusted. Series ID: A2304108K. Source: Australian Bureau of Statistics, Catalogue number 5206.0, Table 2.

Gross Domestic Product: Gross Domestic Product, chain volume measure, $ million, seasonally adjusted. Series ID: A2304402X. Source: Australian Bureau of Statistics, Catalogue number 5206.0, Table 2.

Population: Estimated Resident Population, Persons. Series ID: A2060842F. Source: Australian Bureau of Statistics, Catalogue number 3101.0, Table 1.

Total Taxes: All taxes paid in Australia, $ millions, seasonally adjusted. Series ID: A2302794K. Source: Australian Bureau of Statistics, Catalogue number 5206.0, **Table 22**.

Nominal Gross Domestic Product: Gross Domestic Product, current prices, $ million, seasonally adjusted. Series ID: A2304418T, Catalogue number 5206.0, Table 3.

Interest rate: Rate on 3-month Bank Accepted Bills/Negotiable certificates of deposits, monthly, per cent per annum. Statistical Table F1.1, Series ID: FIRMMBAB90. Reserve Bank of Australia. The quarterly values are the average of the quarterly three months values.

Inflation: Quarterly change in per cent of Consumer Price Index, all groups, excluding volatile items. Statistical Table G1, Series ID: GCPIXVIQP. Reserve Bank of Australia.

Real trade-weighted Index: Australian dollar trade-weighted exchange rate index adjusted for relative consumer price levels, Quarterly, 1995Q1 = 100. Statistical Table F15, Series ID:FRERTWI. Reserve Bank of Australia.

Unemployment rate: Unemployed Persons as Percentage of the Labour Force, per cent, seasonally adjusted. Average of quarterly three months. Series ID: A84423050A. Source: Australian Bureau of Statistics, Catalogue number 6206.0, Table 1.

World industrial production index: Index constructed by Baumeister and Hamilton (2019), monthly. We take the average of the quarterly three months as the quarterly value.